

Adding secrets for private image pull

✓ 9. Pulling Private Images from Docker Hub using Kubernetes Secrets

In production environments, container images are often stored in PRIVATE Docker Hub repositories.

To allow Kubernetes to pull these images securely, we use an `imagePullSecret`.

This section enhances the lab by adding secure authentication to Docker Hub.

✓ 9.1 Create Docker Hub Secret

Replace the values with your actual Docker Hub credentials:

```
kubectl create secret docker-registry dockerhub-secret \
  --docker-server=https://index.docker.io/v1/ \
  --docker-username=<your-dockerhub-username> \
  --docker-password=<your-dockerhub-password> \
  --docker-email=<your-email>
```

✓ Verify the secret:

```
kubectl get secret dockerhub-secret
```

✓ 9.2 Update Deployment YAML to Use `imagePullSecrets`

📄 Updated `nginx-env-demo-deployment.yaml`

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-env-demo
  labels:
    app: nginx-env-demo
spec:
  replicas: 2
  selector:
    matchLabels:
```

```

    app: nginx-env-demo
template:
  metadata:
    labels:
      app: nginx-env-demo
  spec:
    imagePullSecrets:
      - name: dockerhub-secret
    containers:
      - name: nginx
        image: mydockerid/nginx-env-demo:v1
        ports:
          - containerPort: 80
        env:
          - name: WELCOME_MSG
            valueFrom:
              configMapKeyRef:
                name: nginx-env-demo-config
                key: WELCOME_MSG
          - name: ENV
            valueFrom:
              configMapKeyRef:
                name: nginx-env-demo-config
                key: ENV

```

✓ 9.3 Apply the Updated Deployment

```
$ kubectl apply -f nginx-env-demo-deployment.yaml
```

✓ 9.4 Validate Image Pull Authentication

```
$ kubectl describe pod <pod-name> | grep -i image
```

✓ If there are NO ImagePullBackOff or ErrImagePull errors, your secret is working correctly.

✓ 9.5 Production Security Best Practices

- ✓ Never store Docker credentials directly in YAML
- ✓ Always use Secrets
- ✓ Rotate credentials regularly
- ✓ Use RBAC to restrict secret access
- ✓ Store secrets securely in vaults (HashiCorp Vault, AWS Secrets Manager, Azure Key Vault)